

Product Requirements Document (PRD)

AI-Powered Enterprise Data Query Platform

“Chat with DB”

1. Executive Summary

Product Overview

“Chat with DB” is an enterprise-grade AI-powered analytics platform that enables business users to retrieve insights from enterprise databases using plain English instead of SQL.

The platform converts natural language queries into validated SQL queries, executes them securely against Amazon Redshift, and returns results as:

- Human-readable insights
- Tables
- Charts
- Operational summaries

The system is designed primarily for enterprise medical/pharma operational analytics teams where non-technical users frequently depend on data engineering or BI teams for reporting and insights.

Business Opportunity

Modern enterprises suffer from:

- Heavy dependency on data teams
- Slow report turnaround
- SQL skill gaps among business users
- Fragmented analytics workflows
- Delayed operational decision-making

This platform democratizes enterprise data access while maintaining governance, security, and auditability.

Core Value Proposition

For Business Users

- Ask questions in plain English
- Get instant operational insights
- No SQL knowledge required

For Enterprises

- Reduce analytics bottlenecks
- Increase operational agility
- Improve productivity
- Maintain secure governed access

For Data Teams

- Reduce repetitive reporting requests
 - Focus on strategic engineering work
 - Standardize business metric definitions
-

Vision Statement

Build an enterprise AI analytics assistant that becomes the default interface between business users and enterprise data systems.

High-Level Solution Summary

User Query Flow:

Business User

Natural Language Query

AI Query Translation Engine

SQL Validation Layer

Redshift Query Execution

Result Formatter + Insight Generator

Human-Friendly Response + Charts

2. Product Background

Industry Context

Enterprises increasingly rely on:

- Data warehouses
- Business intelligence
- Self-service analytics
- AI-powered automation

However, analytics access remains constrained due to technical barriers.

Existing Market Problems

Current State

Business teams depend on:

- SQL analysts
- BI dashboards
- Static reports
- Data engineering teams

Common Pain Points

- Slow turnaround
 - Report backlog
 - Technical dependency
 - Inconsistent KPI definitions
 - Lack of real-time exploration
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Why This Product Should Exist

The emergence of:

- Large Language Models (LLMs)
- Enterprise AI tooling
- Cloud-native analytics
- Metadata-aware AI systems

creates an opportunity to provide conversational analytics safely within enterprises.

Market Timing Analysis

The timing is favorable because:

- Enterprises are adopting AI rapidly
 - Cloud data warehouses are mainstream
 - Self-service analytics demand is increasing
 - Operational teams require faster decisions
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3. Problem Statement

User Pain Points

Business Users

- Cannot write SQL
- Depend on analysts
- Wait days for reports
- Limited visibility into operational data

Data Teams

- Overloaded with repetitive requests
 - Constant ad-hoc query generation
 - Poor prioritization of strategic work
-

Operational Inefficiencies

- Slow reporting cycles
 - Manual dashboard creation
 - KPI inconsistency
 - Duplicate analytical work
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Market Gaps

Most BI tools:

- Require dashboard creation
 - Require SQL knowledge
 - Lack conversational querying
 - Are difficult for non-technical users
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Technical Problems

- Schema complexity
 - Business terminology mismatch
 - Query safety risks
 - Data governance challenges
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4. Product Vision

Long-Term Vision

Create a conversational enterprise intelligence platform capable of:

- Data exploration
 - Predictive analytics
 - Insight generation
 - Automated reporting
 - Decision intelligence
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Strategic Objectives

- Democratize data access
 - Reduce analytics dependency
 - Standardize enterprise metrics
 - Enable AI-driven operations
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Customer Transformation Goals

Transform:

“Ask analyst for report”

“Ask AI directly”

Expected Business Impact

- Faster decisions
 - Reduced operational delays
 - Higher productivity
 - Lower analytics overhead
-

5. Goals & Objectives

Business Goals

Goal	Target
Reduce reporting dependency	60%
Improve query turnaround	<10 seconds
Increase analytics adoption	3x
Reduce manual reporting effort	50%

Product Goals

- Conversational analytics
 - Secure enterprise querying
 - Human-readable insights
 - Scalable architecture
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User Goals

Users should:

- Ask questions naturally
 - Understand insights easily
 - Trust AI-generated results
-

Success Metrics

KPI	Target
Query success rate	>95%
Query accuracy	>90%
User satisfaction	>4.5/5
Avg response time	<5 sec
AI hallucination rate	<2%

6. Target Audience

Persona 1 — Business Operations Manager

Characteristics

- Non-technical
- Uses Excel heavily
- Needs quick insights

Pain Points

- Waiting for reports
- Manual data analysis

Goals

- Fast answers
 - Operational visibility
-

Persona 2 — Pharma Marketing Analyst

Characteristics

- Moderate analytics exposure
- Understands metrics
- Limited SQL knowledge

Goals

- Campaign performance insights
 - Country-level analytics
-

Persona 3 — Enterprise Admin

Goals

- Governance
 - Access control
 - Auditability
 - Compliance
-

7. User Stories

Story 1

As a business user, I want to ask questions in plain English, so that I can get insights without SQL knowledge.

Acceptance Criteria

- User can submit natural language query
- AI generates SQL
- Results are displayed within 10 seconds

Priority

Must Have

Story 2

As an admin, I want all queries logged, so that audit requirements are met.

Acceptance Criteria

- Query stored with timestamp
 - User identity stored
 - SQL stored securely
-

8. Use Cases

Primary Use Cases

- Sales analytics
- Campaign performance
- Product performance
- Expiry tracking

- Country analytics

Secondary Use Cases

- Trend analysis
 - KPI monitoring
 - Export reports
-

Failure Scenarios

Scenario	System Response
Invalid query	Friendly error
AI hallucination	Validation rejection
Timeout	Retry suggestion
Unauthorized table access	Access denied

9. Product Features

Feature: Natural Language Querying

Description

Users type business questions in plain English.

Business Impact

Reduces dependency on technical teams.

Priority

Must Have

Feature: AI SQL Generation

Description

AI converts text into optimized SQL.

Technical Considerations

- Schema-aware prompting
 - Business terminology mapping
 - SQL validation
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Feature: Query Validation Engine

Responsibilities

- Prevent destructive SQL
 - Enforce read-only queries
 - Limit row counts
 - Detect unsafe operations
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Feature: Insight Generation

Capabilities

- Human-readable summaries
 - KPI explanations
 - Trend interpretation
-

Feature: Query History

Capabilities

- Saved prompts
 - Recent queries
 - Re-run queries
-

Feature: Visualization Engine

Charts

- Bar charts
 - Pie charts
 - Trend graphs
 - KPI cards
-

Feature: Audit Logging

Captured Data

- User
 - Query
 - SQL
 - Timestamp
 - Execution duration
-

10. Functional Requirements

Query Processing

Inputs

- Natural language query

Outputs

- SQL query
 - Data results
 - AI-generated insights
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Validation Rules

The system must:

- Allow SELECT only
 - Reject UPDATE/DELETE/DROP
 - Restrict table access
 - Enforce query timeout
-

Workflow

```
graph TD; A[User Query] --> B[AI Conversion]; B --> C[SQL Validation]; C --> D[Redshift Execution]; D --> E[Result Processing]; E --> F[Insight Generation]; F --> G[UI Rendering];
```

User Query

AI Conversion

SQL Validation

Redshift Execution

Result Processing

Insight Generation

UI Rendering

Permissions

Role	Permissions
User	Query approved datasets
Admin	Manage permissions
Super Admin	System configuration

11. Non-Functional Requirements

Scalability

System must support:

- 10,000+ users
 - Concurrent query execution
 - Horizontal API scaling
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Reliability

- 99.9% uptime
 - Automatic retries
 - Graceful degradation
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Performance

Metric	Target
Query response	<5 sec
AI generation	<3 sec

Security

- Encryption at rest
 - TLS in transit
 - IAM-based access
 - Query sanitization
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Observability

- CloudWatch logs
 - Query tracing
 - Metrics dashboards
 - Error alerts
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12. UX & UI Requirements

UX Principles

- Minimal learning curve
 - Conversational interaction
 - Enterprise simplicity
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Navigation

Dashboard

Query Workspace

Saved Queries

Query History

Insights

Admin Panel

UI Components

- Chat interface
- SQL preview
- Data tables
- Chart cards
- Export actions

Accessibility

- WCAG 2.1 compliance
- Keyboard navigation
- Screen reader support

13. Information Architecture

Data Structure

Workspace

Queries

Saved Reports

Insights

Audit Logs

Search Behavior

- Semantic query search
 - Saved prompt search
 - Query history filtering
-

14. Technical Architecture

Frontend Stack

Layer	Technology
Framework	Next.js
Styling	Tailwind CSS
Components	Shadcn UI
State Mgmt	Zustand/Redux
Charts	Recharts

Backend Stack

Layer	Technology
API	API Gateway
Compute	AWS Lambda
Database	Amazon Redshift
Secrets	AWS Secrets Manager
Logging	CloudWatch

AI Layer

Responsibilities

- Text-to-SQL conversion
- Prompt orchestration
- Schema injection
- SQL optimization

Infrastructure Flow



15. Data Model & Database Requirements

Core Entities

Entity	Purpose
User	Authentication
Query	Query history
AuditLog	Compliance
SavedPrompt	Reusable prompts

Audit Requirements

Store:

- User activity
 - Query text
 - SQL text
 - Response metadata
-

16. API Requirements

POST /query

Request

```
{  
  "query": "Show top products by country"  
}
```

Response

```
{  
  "sql": "SELECT ...",  
  "summary": "Top products are...",  
  "results": []  
}
```

Authentication

- JWT

- SSO integration
- IAM federation

Rate Limiting

User Type	Limit
Standard	60 req/min
Admin	120 req/min

17. Security & Compliance

Security Controls

- RBAC
 - Table-level access
 - Column masking
 - SQL validation
-

Compliance

Potential future support:

- HIPAA
 - SOC2
 - GDPR
-

Threat Mitigation

Threat	Mitigation
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SQL injection Query parser
Prompt injection Prompt filtering
Data leakage Access policies

18. Integrations

Planned Integrations

Integration	Purpose
Slack	Insight sharing
Email	Scheduled reports
Power BI	Export support
Tableau	Visualization sync

19. Analytics & Tracking

Events

- Query submitted
 - Query success
 - Query failure
 - Saved prompt creation
-

Dashboards

- Usage metrics
- Query latency
- AI accuracy
- User engagement

20. AI/Automation Requirements

AI Use Cases

- Text-to-SQL
 - Insight summarization
 - KPI interpretation
 - Chart recommendations
-

AI Safety

- Schema grounding
 - Confidence scoring
 - Query validation
 - Hallucination detection
-

Prompt Workflow

User Query
Schema Context Injection
Business Vocabulary Injection
AI Generation
SQL Validation

21. Monetization Strategy

Internal Enterprise Product

Initial deployment:

- Internal-only
 - Productivity optimization
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Future SaaS Potential

Possible pricing:

- Per-seat licensing
 - Query-based pricing
 - Enterprise subscription
-

22. Competitive Analysis

Competitor	Weakness
Traditional BI tools	Complex
SQL IDEs	Technical users only
Generic AI chatbots	No governance

Competitive Advantage

- Enterprise-safe AI querying
 - Business terminology understanding
 - Secure SQL governance
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23. Go-To-Market Strategy

Initial Rollout

Phase 1:

- Internal pilot users
 - Limited datasets
 - Controlled rollout
-

Adoption Strategy

- Training sessions
 - Internal champions
 - Feedback loops
-

24. Roadmap

MVP

Scope

- 3 tables
 - Read-only queries
 - Basic charts
 - Query history
-

Phase 2

- Multi-schema support
- AI explanations

- Saved dashboards
 - Scheduled reports
-

Phase 3

- Predictive analytics
 - AI recommendations
 - Conversational memory
 - Cross-database querying
-

25. Risks & Assumptions

Risks

Risk	Severity
AI hallucination	High
Unauthorized access	High
Query cost spikes	Medium

Mitigation

- SQL validation
 - Row limits
 - Access policies
 - Query sandboxing
-

26. QA & Testing Strategy

Testing Types

- Unit testing
 - API testing
 - E2E testing
 - Load testing
 - Security testing
-

UAT

Business users validate:

- Query correctness
 - Insight usefulness
 - KPI accuracy
-

27. Deployment & DevOps

Environment Strategy

Environment	Purpose
Dev	Development
QA	Testing
Staging	Pre-production
Prod	Production

CI/CD

GitHub
Build
Test
Deploy
Monitoring

Monitoring

- CloudWatch
 - Error alerts
 - Query latency monitoring
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28. Success Criteria

Launch Readiness

- Query accuracy >90%
 - Response time <5 sec
 - Zero destructive SQL allowed
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Product Success

- Increased analytics adoption
 - Reduced reporting dependency
 - Positive user feedback
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29. Appendices

Glossary

Term	Meaning
NLP	Natural Language Processing
RBAC	Role-Based Access Control
KPI	Key Performance Indicator

Assumptions

- Users have enterprise login access
 - Redshift contains clean structured data
 - AI service has schema awareness
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Open Questions

- Multi-language support?
- Multi-database support?
- Real-time streaming analytics?
- Fine-grained semantic layer management?